Find A Gene Project (Final)

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Q1. Tell me the name of a protein you are interested in. Include the species and the accession number. This can be a human protein or a protein from any other species as long as it's function is known.

Name: kinesin family member 11

Accession: NP_004514

Species: Homo sapiens 1

Function: Enables ATP, microtubule, nucleotide, protein, and protein kinase binding. It also enables microtubule and plus-end-directed microtubule motor activities.

Q2. Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search against Mus musculus (House mouse) ESTs Database: Expressed Sequence Tags (est)

Organism: Mus musculus (taxid:10090)

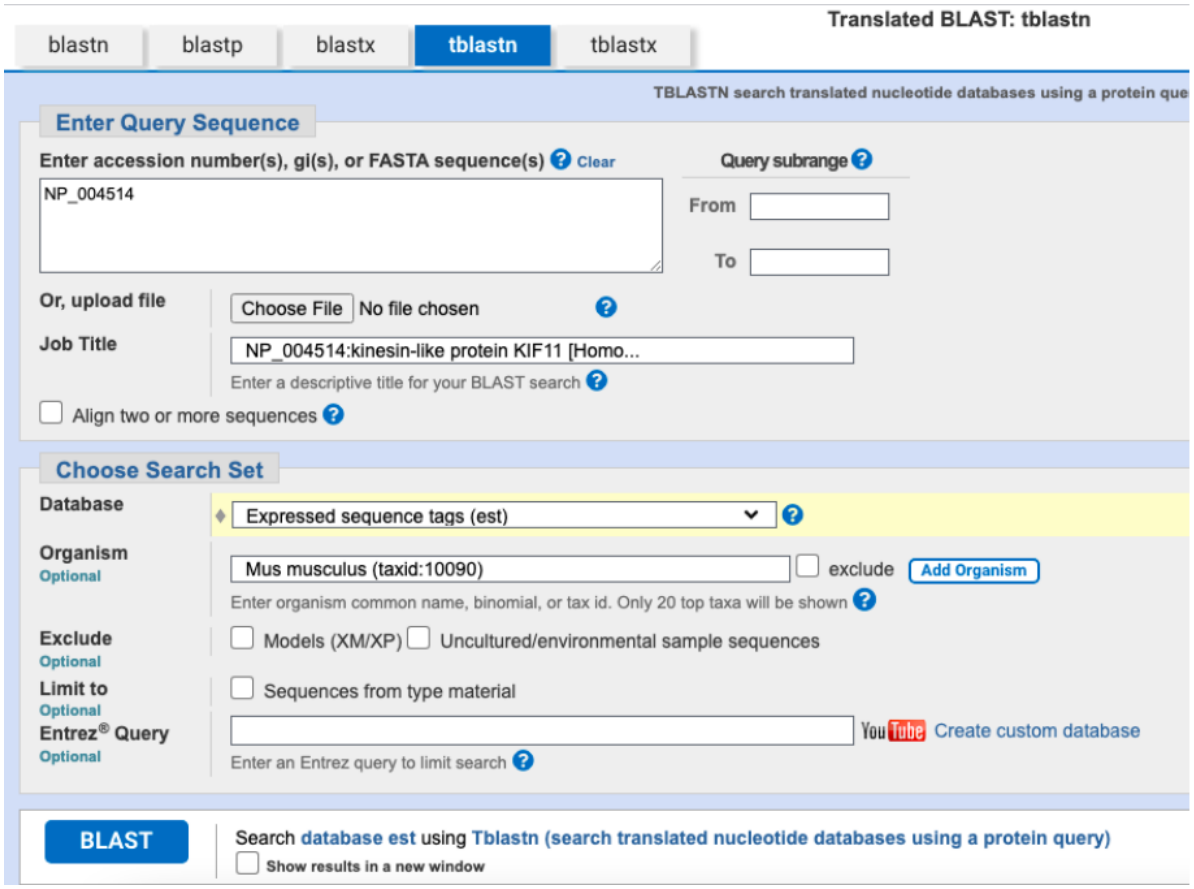
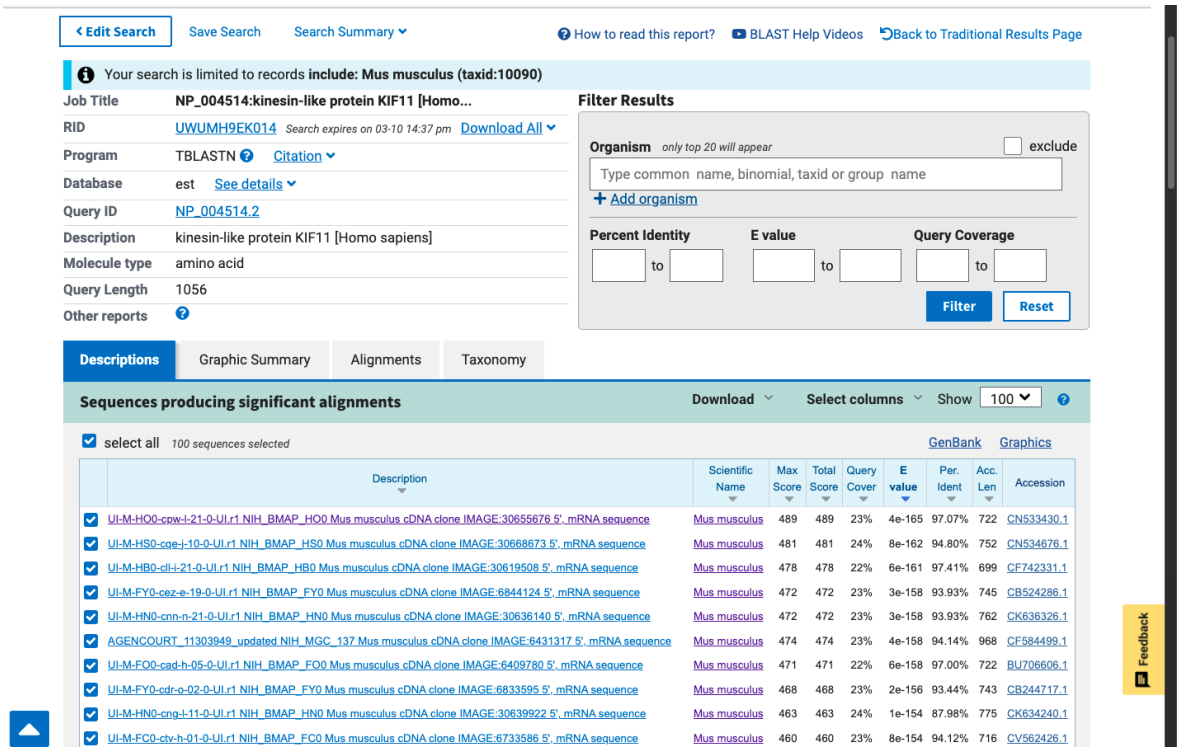


Figure 1: Blast Method Used



Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
UI-M-HO0-cpw-l-21-0-UI.r1 NIH_BMAP_HO0 Mus musculus cDNA clone IMAGE:30655676 5', mRNA sequence	Mus musculus	489	489	23%	4e-165	97.07%	722	CN533430.1
UI-M-HS0-cqe-j-10-0-UI.r1 NIH_BMAP_HS0 Mus musculus cDNA clone IMAGE:30668673 5', mRNA sequence	Mus musculus	481	481	24%	8e-162	94.80%	752	CN534676.1
UI-M-HB0-cll-i-21-0-UI.r1 NIH_BMAP_HB0 Mus musculus cDNA clone IMAGE:30619508 5', mRNA sequence	Mus musculus	478	478	22%	6e-161	97.41%	699	CF742331.1
UI-M-FY0-cez-e-19-0-UI.r1 NIH_BMAP_FY0 Mus musculus cDNA clone IMAGE:6844124 5', mRNA sequence	Mus musculus	472	472	23%	3e-158	93.93%	745	CB524286.1
UI-M-HN0-cnn-n-21-0-UI.r1 NIH_BMAP_HN0 Mus musculus cDNA clone IMAGE:30636140 5', mRNA sequence	Mus musculus	472	472	23%	3e-158	93.93%	762	CK636326.1
AGENCOURT_11303949_updated NIH_MGC_137 Mus musculus cDNA clone IMAGE:6431317 5', mRNA sequence	Mus musculus	474	474	23%	4e-158	94.14%	968	CF584499.1
UI-M-FO0-cad-h-05-0-UI.r1 NIH_BMAP_FO0 Mus musculus cDNA clone IMAGE:6409780 5', mRNA sequence	Mus musculus	471	471	22%	6e-158	97.00%	722	BU706606.1
UI-M-FY0-cdr-o-02-0-UI.r1 NIH_BMAP_FY0 Mus musculus cDNA clone IMAGE:6833595 5', mRNA sequence	Mus musculus	468	468	23%	2e-156	93.44%	743	CB244717.1
UI-M-HN0-cng-l-11-0-UI.r1 NIH_BMAP_HN0 Mus musculus cDNA clone IMAGE:30639922 5', mRNA sequence	Mus musculus	463	463	24%	1e-154	87.98%	775	CK634240.1
UI-M-FC0-ctv-h-01-0-UI.r1 NIH_BMAP_FC0 Mus musculus cDNA clone IMAGE:6733586 5', mRNA sequence	Mus musculus	460	460	23%	8e-154	94.12%	716	CV562426.1

Figure 2: Search Output List Top Hits

Chosen Match: Accession CN533430.1, a 722 base pair clone from Mus musculus.

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UI-M-HO0-cpw-l-21-0-UI.r1 NIH_BMAP_HO0 Mus musculus cDNA clone IMAGE:30655676 5', mRNA sequence

Sequence ID: CN533430.1 Length: 722 Number of Matches: 1

Range 1: 6 to 722 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
489 bits(1260)	4e-165	Compositional matrix adjust.	232/239(97%)	235/239(98%)	0/239(0%)	+3

```
Query  47   RKEVSVRTGGLADKSSRKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQ  106
            RKEVSVRT GL DK+S+KTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQ
Sbjct  6    RKEVSVRTAGLTDKTSKKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQ  185

Query  107  TGTGKTFTMEGERSPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNE  166
            TGTGKTFTMEGERSPNE YTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNE
Sbjct  186  TGTGKTFTMEGERSPNEVYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNE  365

Query  167  ELFDLLNPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAAT  226
            ELFDLL+PSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAAT
Sbjct  366  ELFDLLSPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAAT  545

Query  227  LMNAYSSRSHSVFSVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREA  285
            LMNAYSSRSHSVFSVTIHMK TTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREA
Sbjct  546  LMNAYSSRSHSVFSVTIHMKXTTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREA  722
```

Figure 3: Chosen Match

Q3. Gather information about this "novel" protein. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

Chosen Sequence (FASTA Format): >CN533430.1 UI-M-HO0-cpw-l-21-0-UI.r1 NIH_BMAP_HO0 Mus musculus cDNA clone IMAGE:30655676 5', mRNA sequence

RKEVSVRTAGLTDKTSKKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTI-
FAYGQTGTGKTFTMEGERSPNEVYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVS-
LLEIYNEELFDLLSPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEK-
GAAKRTTAATLMNAYSSRSHSVFSVTIHMKXTTIDGEELVKIGKLNLVDLAGSENIGRS-
GAVDKRAREA

Protein Name: UI-M-HO0-cpw-l-21-0-UI.r1 NIH_BMAP_HO0 Mus musculus cDNA clone IMAGE:30655676 5', mRNA sequence

Species: Mus musculus Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Glires; Rodentia; Myomorpha; Muroidea; Muridae; Murinae; Mus; Mus.

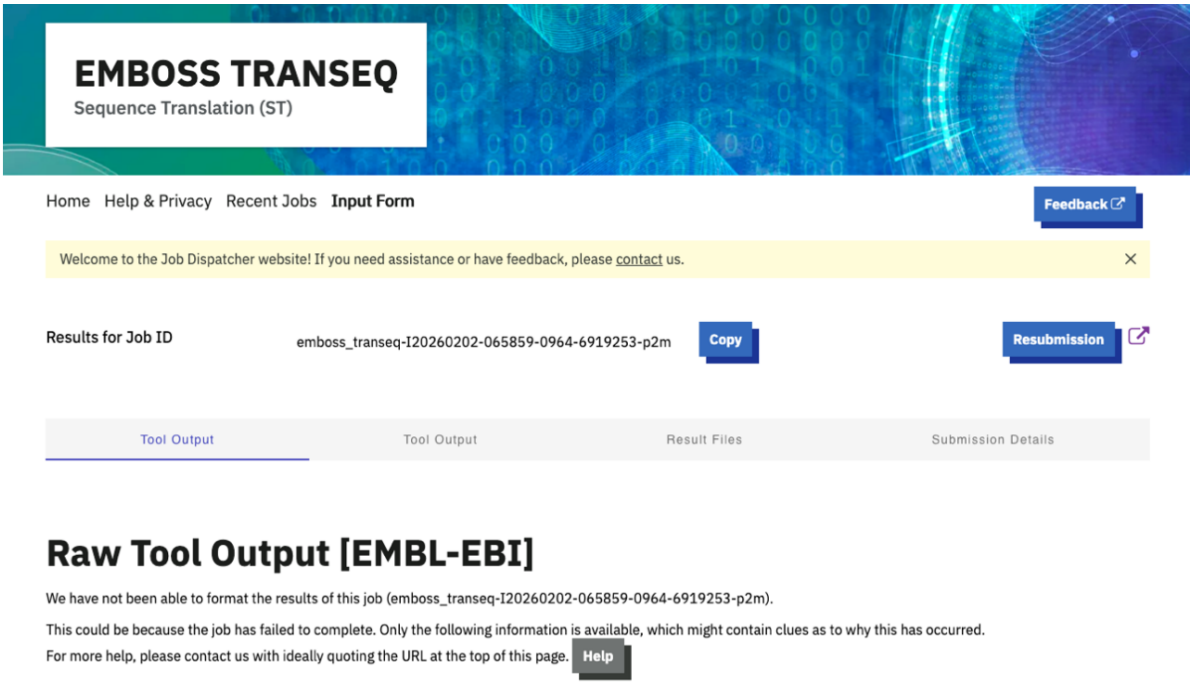


Figure 4: Figure 3. EMBOSS Transeq Search

Q4. Prove that this gene, and its corresponding protein, are novel. For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]), and use it as a query in a blastp search of the nr database at NCBI.

Description: A BLASTP search against NR database yielded a top hit result to a protein from Rattus norvegicus (NP_001162583.1).

Species: Rattus norvegicus

Name: kinesin-like protein KIF11 [Rattus norvegicus] Percent Identity: 99.16%. Since the top match has less than 100% identity, this must mean the protein is novel because there is no identical match that is 100% to this specific species in the protein database.

Sequences producing significant alignments Download Select columns Show 100

select all 100 sequences selected GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
kinesin family member 11, isoform CRA_b, partial [Mus musculus]	Mus musculus	517	517	100%	1e-153	99.58%	1064	EDL41788.1
kinesin-like protein KIF11 [Mus musculus]	Mus musculus	516	516	100%	2e-153	99.58%	1052	NP_034745.1
kinesin-like protein KIF11 [Rattus norvegicus]	Rattus norvegicus	515	515	100%	5e-153	99.16%	1056	NP_001162583.1
kinesin-like protein KIF11 [Rattus rattus]	Rattus rattus	515	515	100%	5e-153	99.16%	1056	XP_032747743.1
rCG48024 [Rattus norvegicus]	Rattus norvegicus	515	515	100%	5e-153	99.16%	850	EDM13198.1
kinesin-like protein KIF11 [Peromyscus maniculatus bairdii]	Peromyscus maniculatus bairdii	514	514	100%	1e-152	98.74%	1055	XP_042118941.2
kinesin-like protein KIF11 [Peromyscus leucopus]	Peromyscus leucopus	514	514	100%	1e-152	98.74%	1055	XP_028738726.1
kinesin-like protein KIF11 isoform X1 [Peromyscus maniculatus bairdii]	Peromyscus maniculatus bairdii	514	514	100%	2e-152	98.74%	1055	XP_042118941.1
kinesin-like protein KIF11 [Arvicanthis niloticus]	Arvicanthis niloticus	514	514	100%	2e-152	99.16%	1054	XP_034350713.1
kinesin-like protein KIF11 [Mastomys coucha]	Mastomys coucha	513	513	100%	2e-152	98.74%	1055	XP_031246053.1
Chain A, KINESIN-LIKE PROTEIN KIF11 [Homo sapiens]	Homo sapiens	500	500	100%	5e-152	97.07%	348	3ZCW_A
Chain A, Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	503	503	100%	5e-152	97.07%	382	3WPN_A
Chain A, Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	100%	5e-152	97.07%	367	1Q0B_A
kinesin-like protein KIF11 isoform X1 [Phodopus roborovskii]	Phodopus roborovskii	512	512	100%	5e-152	98.33%	1055	XP_051055532.1
Chain A, Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	100%	5e-152	97.07%	367	5ZO7_A
Chain A, Kinesin-like protein KIF11 [Homo sapiens]	Homo sapiens	501	501	100%	5e-152	97.07%	367	6TLE_A
Chain A, KINESIN-RELATED MOTOR PROTEIN Eg5 [Homo sapiens]	Homo sapiens	501	501	100%	5e-152	97.07%	368	1II6_A

Figure 5: BLASTP Search Against NR Dataset

Q5. Generate a multiple sequence alignment with your novel protein, your original

query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width.

Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled sequences for alignment: 1. Human_KLP | 7484953 | ref | NP_004514 | kinesin-like protein KIF11 [Homo sapiens]

VQENIQQKSKDIVNKMTFHSQKFCADSDGFSQELRNFNQEGTKLVEESVKHSDKLNGN-
LEKISQETEQRCESLNTRTVYFSEQWVSSLNEREQELHNLLEVVSQCCEASSSDITEKS-
DGRKAAHEKQHNIFLDQMTIDEDKLIAQNLELNETIKIGLTKLNCFLEQDLKLDIPT-
GTTPQRKSYLYPSTLVRTEPREHLLDQLKRKQPELLMMLNCSENNKEETIPDVD-
VEEAVLGQYTEEPLSQEPSVDAGVDCSSIGGVPFFQHKKSHGKDKE

2. Mus musculus_KLP | 30655676 | ref | CN533430.1 | kinesin-like protein KIF11 [Homo sapiens]

RKEVSVRTAGLTDKTSKKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTI-
FAYGQTGTGKTFTMEGERSPNEVYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVS-
LLEIYNEELFDLLSPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEK-
GAAKRTTAATLMNAYSSRSHSVFSVTIHMKXTTIDGEELVKIGKLNLVDLAGSENIGRS-
GAVDKRAREA

3. Pig gi | ref | DN125747.1 | 1139617 MARC 4PIG Sus scrofa cDNA 3'

GKTFTMEGERSPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLL-
NPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLM-
NAYSSRSHSVFSVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGN-
INQSLLTLGRVITALVERTPHVPYRESKLTRILQDSLGGRTRTSIIATISPAslnleetlst-
leYAHRAKNILNKPEVNQKLTKKALIKEYTEEIERLKRDLAAARE

4. Tropical_clawed_frog gi | 7695825 | ref | CX924637.2 | JGI_CAAN10013.fwd NIH_XGC_tropTe4 Xenopus tropicalis

RKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQTGTGKTFT-
MEGERSPNEEYTWEEDPLAGIIPRTLHQIFEKLTDNGTEFSVKVSLLEIYNEELFDLL-
NPSSDVSERLQMFDDPRNKRGVIIKGLEEITVHNKDEVYQILEKGAAKRTTAATLM-
NAYSSRSHSVFSVTIHMKETTIDGEELVKIGKLNLVDLAGSENIGRSGAVDKRAREAGNIN-
QSLLTLGRVITALVERTPHVPYRESKLTRILQDSL

5. Domestic_cattle gi | | ref | HX934525.1 | HX934525 Muscle satellite cell of Korean cattle (Hanwoo), normalized cDNA library Bos taurus

YAHRAKNILNKPEVNQKLTKKALIKEYTEEIERLKRDLAAAREKNGVYISEENFRVMS-
GKLTVQEEQIVELIEKIGAVEEELNRVTELFMDNKNELDQCKSDLQNKTQELET-
TQKHLQETKLQLVKEEYITSALESTEEKLHDAASKLLNTVEETTKDVSGLHSKL-
DRKKAVDQHNAEAQDIFGKNLNSLFNNMEELIKDGSSKQKAMLEVHKTLFGN-
LLSSSVSALDTITTVALGSLTSIPENVSTHVSQIFNMILKEQSLAAESKTVL

Alignment: Obtained using MUSCLE (version 3.8) at EBI

CLUSTAL multiple sequence alignment by MUSCLE (3.8)

```
Domestic_cattle          ----------------YAHRAKNILNKPEVNQKLTKKALIKEYTEEIERLKRDLAAAREK
Human_KLP                ------------VQENIQQKSKDIVNK---------------------MTFHSQKFCADS
Mus                      RKEVSVRTAGLTDKTSKKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQ
Pig                      ------------------------------------------------------------
Tropical_clawed_frog     ----------------RKTYTFDMVFGASTKQIDVYRSVVCPILDEVIMGYNCTIFAYGQ
```

```
Domestic_cattle          NGVYISEENFRVMSGKLTVQEEQIVE--LIEKIGAVEEELNRVTELFMDNKNELDQC-KS
Human_KLP                DGFSQELRNFN-QEGTKLVEESVKHSDKLNGNLEKISQETEQRCESLNTRTVYFSEQWVS
Mus                      TGTG---KTFT-MEGERSPNEVYTWEEDPLAGI--IPRTLHQIFEKLTDNGTEFSVK-VS
Pig                      ---G---KTFT-MEGERSPNEEYTWEEDPLAGI--IPRTLHQIFEKLTDNGTEFSVK-VS
Tropical_clawed_frog     TGTG---KTFT-MEGERSPNEEYTWEEDPLAGI--IPRTLHQIFEKLTDNGTEFSVK-VS
                                 .*   .*    :*    .      :  :     .  * :  .   :.    *

Domestic_cattle          DLQNKTQELETTQKHLQETKLQLVKEEYITSALESTEEKLHDAASKLLNTVEE-TTKDVS
Human_KLP                SLNEREQELHNLLEVVSQCCEASSSD--ITEKSDGRKAA-HEKQHNIF--LDQMTIDEDK
Mus                      LLEIYNEELFDLLS--------PSSD--VSERLQMFDDP-RNKRGVIIKGLEEITVHNKD
Pig                      LLEIYNEELFDLLN--------PSSD--VSERLQMFDDP-RNKRGVIIKGLEEITVHNKD
Tropical_clawed_frog     LLEIYNEELFDLLN--------PSSD--VSERLQMFDDP-RNKRGVIIKGLEEITVHNKD
                          *:   :**    .          .:  ::.  :  .   .:    ::  ::: *  : .

Domestic_cattle          GLHSKLDRKKAVDQHNAEAQDIF--------------GKNLNSLFN------NMEELIKD
Human_KLP                LIAQNLELNETIKI-GLTKLNCFLEQDLKLDIPTGTTPQRKSYLYPSTLVRTEPREHLLD
Mus                      EVYQILEKGAAKRTTAATLMNAY--------------SSRSHSVFSVTI---HMKXTTID
Pig                      EVYQILEKGAAKRTTAATLMNAY--------------SSRSHSVFSVTI---HMKETTID
Tropical_clawed_frog     EVYQILEKGAAKRTTAATLMNAY--------------SSRSHSVFSVTI---HMKETTID
                          : . *:   :         : :               ..   ::             *

Domestic_cattle          GSSKQKAMLEVHKTLFGNLLSSSVSALDTITTVALGSLTSIPENVSTHVSQIFNMILKEQ
Human_KLP                QLKRKQPELLMMLNCSENNKEETIPDVD-VEEAVLGQYTEEPLSQEPSVDAGVDCSSIGG
Mus                      GEELVKIGKLNLVDLAGSENIGRSGAVD-KRAREA------------------------
Pig                      GEELVKIGKLNLVDLAGSENIGRSGAVD-KRAREAGNINQSLLTLGRVITALVERTP--H
Tropical_clawed_frog     GEELVKIGKLNLVDLAGSENIGRSGAVD-KRAREAGNINQSLLTLGRVITALVERTP--H
                           .  :           .        :*

Domestic_cattle          SLAAESKTVL--------------------------------------------------
Human_KLP                VPFFQHKKSHGKDKE---------------------------------------------
Mus                      ------------------------------------------------------------
Pig                      VPYRESKLTRILQDSLGGRTRTSIIATISPASLNLEETLSTLEYAHRAKNILNKPEVNQK
Tropical_clawed_frog     VPYRESKLTRILQDSL--------------------------------------------


Domestic_cattle          --------------------------
Human_KLP                --------------------------
Mus                      --------------------------
Pig                      LTKKALIKEYTEEIERLKRDLAAARE
Tropical_clawed_frog     --------------------------
```

Q6. Create a phylogenetic tree, using either a parsimony or distance-based approach.

Bootstrapping and tree rooting are optional. Use "simple phylogeny" online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

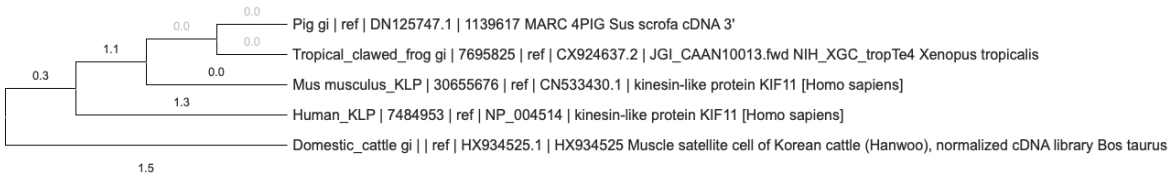


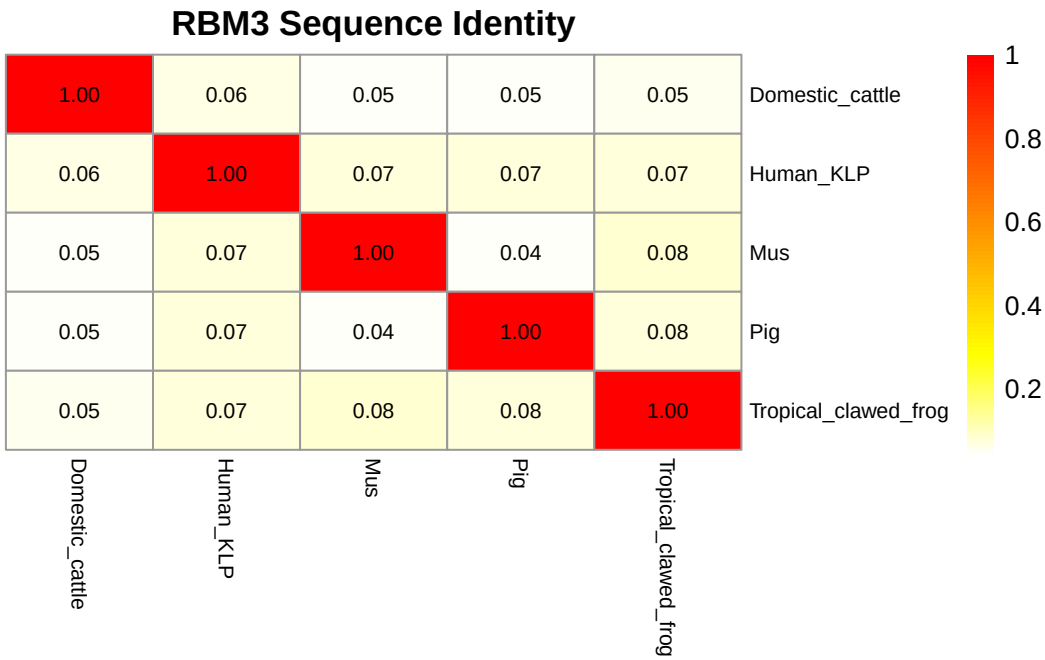
Figure 6: Phylogenetic tree

Q7. Generate a sequence identity based heatmap of your aligned sequences using R.

If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and "Save as" FASTA format for example). Read this FASTA format alignment into R with the help of functions in the Bio3D package. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

```
library(bio3d)
library(pheatmap)

aln <- read.fasta("Q7_Heatmap.fas")
ide_matrix <- seqidentity(aln)

pheatmap(
  ide_matrix,
  main = "RBM3 Sequence Identity",

  cluster_rows = FALSE,
  cluster_cols = FALSE,
  fontsize = 10,

  color = rev(heat.colors(100)),

  display_numbers = TRUE,
```

```
  number_format = "%.2f",
  number_color = "black",
  fontsize_number = 8,
  fontsize_row = 8,
  fontsize_col = 8,
)
```



Q8. Using R/Bio3D (or an online blast server if you prefer), search the main protein

structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 unique hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

```
library(bio3d)

# Extract the first sequence (row 1) from the alignment object 'aln'. 'aln$ali' is a matrix o
seq_vec <- aln$ali[1, ]
```

```r
# Collapse the vector of characters into a single string sequence and remove gap characters
seq_str <- paste(seq_vec, collapse = "")
seq_str <- gsub("-", "", seq_str)

# Print the length of the ungapped query sequence and the first 50 residues of the query sequ
cat("Query length:", nchar(seq_str), "\n")
```

```
Query length: 270
```

```r
cat("Query preview:", substr(seq_str, 1, 50), "...\n")
```

```
Query preview: YAHRAKNILNKPEVNQKLTKKALIKEYTEEIERLKRDLAAAREKNGVYIS ...
```

```r
# Run BLAST search against the PDB database using the query sequence
blast_results <- blast.pdb(seq_str)
```

```
 Searching ... please wait (updates every 5 seconds) RID = UX0DRG9K014
 ..
 Reporting 27 hits
```

```r
# Extract the hit table from the BLAST results
hits <- blast_results$hit.tbl

# Derive the 4-character PDB structure ID from the 'pdb.id' field and remove duplicate entrie
hits$structureId <- substr(hits$pdb.id, 1, 4)
unique_hits <- hits[!duplicated(hits$structureId), ]

# Select the top 3 unique PDB hits
top3 <- unique_hits[1:3, ]

# Retrieve annotation (metadata) for the top 3 PDB structure IDs
annot <- pdb.annotate(top3$structureId)

# Ensure the structureId column is present and explicitly named and remove duplicate annotati
annot$structureId <- annot$structureId
annot_unique <- annot[!duplicated(annot$structureId), ]

# Match the order of annotations to the order of 'top3$structureId'. 'match' returns indices
idx <- match(top3$structureId, annot_unique$structureId)
```

```r
# Subset the annotation table so rows align with the top3 hits
annot_sub <- annot_unique[idx, ]

# Make a table combining BLAST metrics and PDB annotations
result_table <- data.frame(
  structureId = top3$structureId,
  experimentalTechnique = annot_sub$experimentalTechnique,
  resolution = annot_sub$resolution,
  source = annot_sub$source,
  Evalue = top3$evalue,
  Identity = top3$identity
)
```

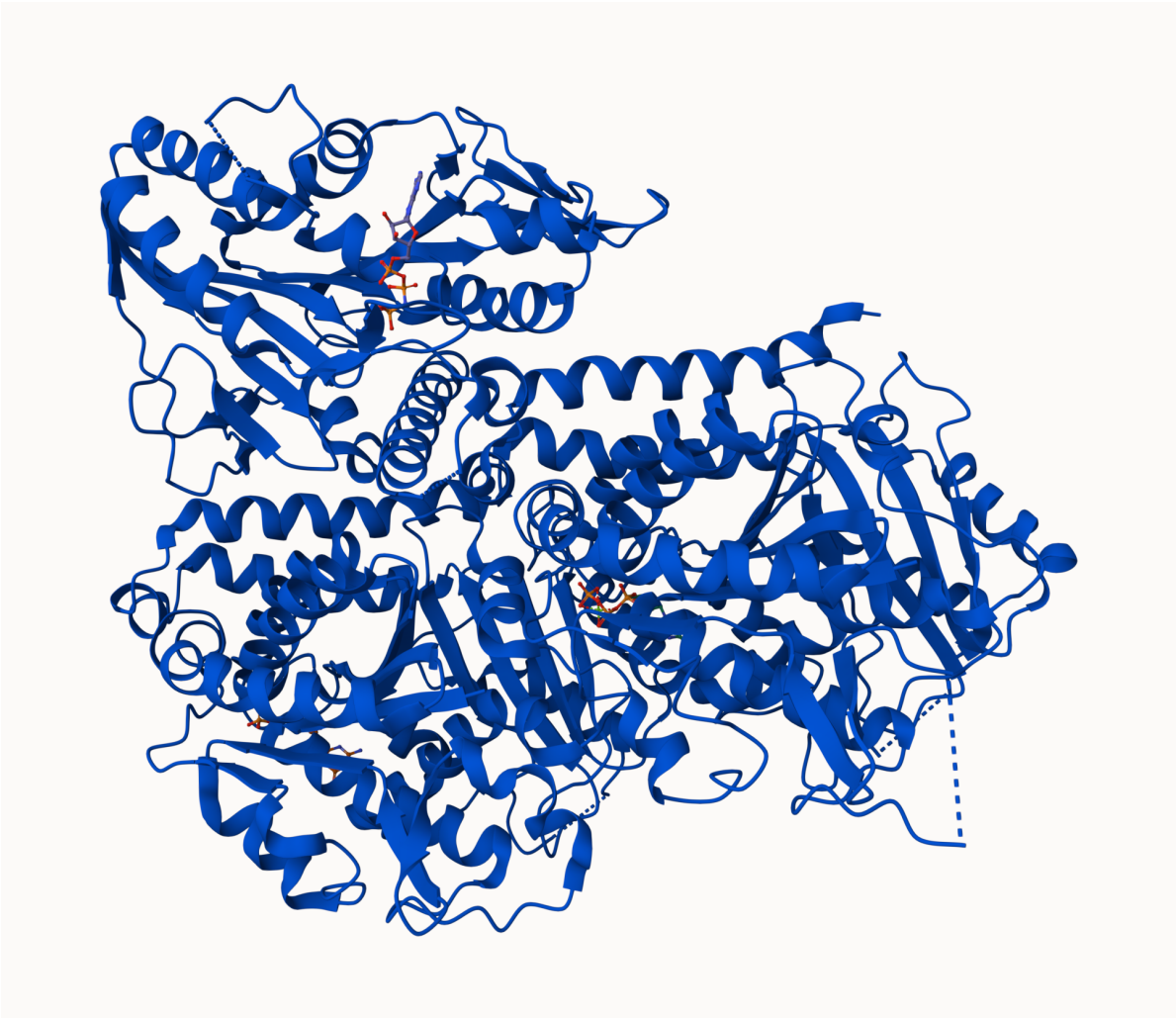
```r
print(result_table)
```

```
  structureId experimentalTechnique resolution                  source   Evalue
1        5JV3                 X-ray      2.006            Homo sapiens 2.36e-
09
2        6VPO                    EM      4.400 Drosophila melanogaster 5.86e-
05
3        7A3Z                 X-ray      2.095  Caenorhabditis elegans 6.08e-
05
  Identity
1   75.000
2   95.238
3   62.857
```

Q9. Generate a molecular figure of one of your identified PDB structures using VMD.

You can optionally highlight conserved residues that are likely to be functional. Please use a white or transparent background for your figure (i.e. not the default black). Based on sequence similarity.

How likely is this structure to be similar to your "novel" protein?



This protein is highly likely to be similar to my "novel" protein because it has a percent identity >90% (e.g. 95%).

Q10

(i) Using your computed structure model (or your closest homologue of known structure from the PDB) predict and locate potential small molecule binding sites using the CASTpFold server (https://cfold.bme.uic.edu/castpfold/). Provide an image or screen-shot of your largest predicted pockets "negative volume" and provide it's area and volume.

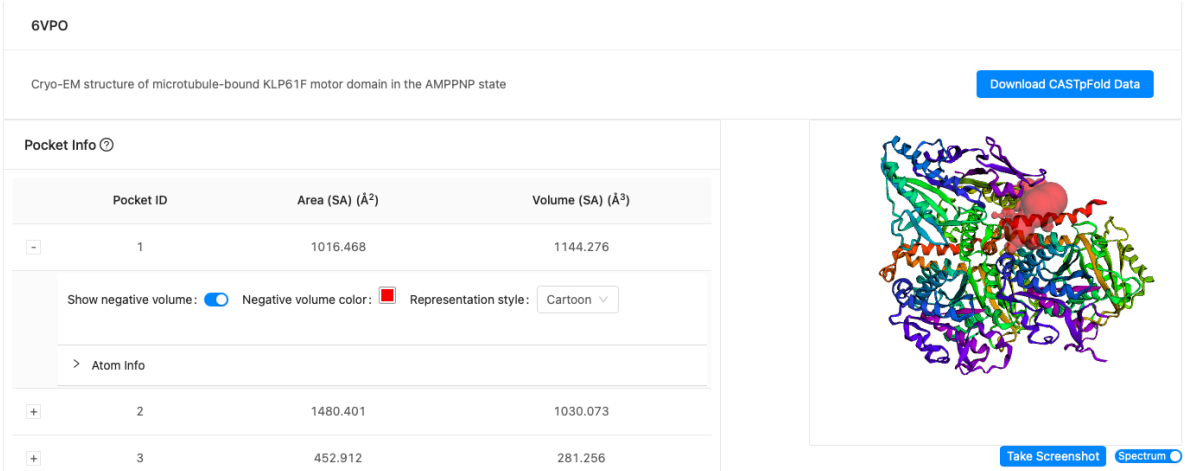


Figure 7: Using CASTpFold Server Largest Predicted Pockets

Area: 1016.468

Volume: 1144.276

(ii) Perform a "Target" search of ChEMBEL (https://www.ebi.ac.uk/chembl/) with your novel sequence. Are there any Target Associated Assays and ligand efficiency data reported that may be useful starting points for exploring potential inhibition of your novel protein?

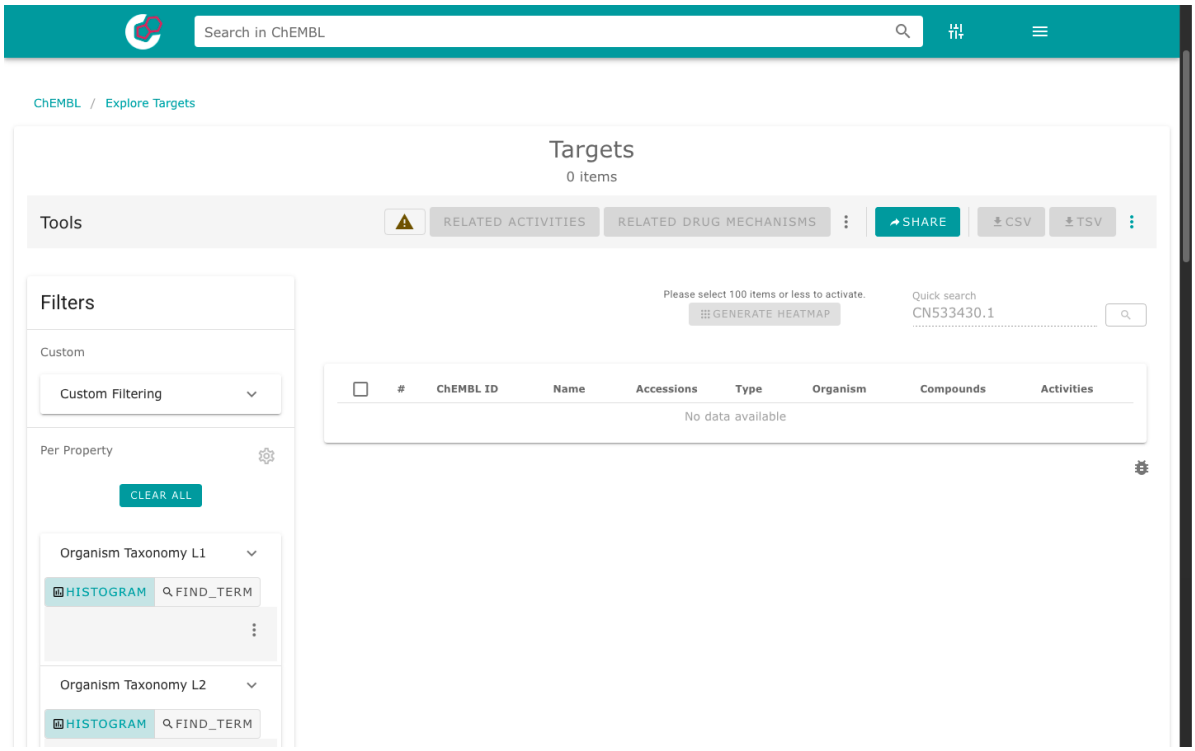


Figure 8: Proof of ChEMBL Target Search with Novel (CN533430.1) Protein

A ChEMBL target search using my novel sequence (CN533430.1) returns no matching targets as of March 8, 2026 because there are no target-associated assays. This means that there are no ligand efficiency metrics (LE, BEI, SEI) reported that could be useful for exploring potential inhibition of CN533430.1. This is the expected result for a globin-like protein.

(iii) Briefly discuss (100 words max) the druggability of your novel protein based on:

- Presence of well-defined pockets (output of tools like CASTpFold),
- Existence of known inhibitors for related proteins (your search of ChEMBEL),
- Conservation of binding sites across homologs (your conservation analysis in Q10),
- Potential therapeutic applications if this protein were targeted (you can use ChatGPT, Claude etc. backed up by your reading of the literature here).

This sequence encodes the kinesin-like motor protein KIF11, a mitotic kinesin. Structural analyses of KIF11 family proteins reveal deep, well-defined ATP-binding and allosteric pockets, but CASTpFold results for the novel variant indicate no additional druggable cavities beyond the conserved motor domain. ChEMBL searches show multiple known KIF11 inhibitors, but none map uniquely to CN533430.1. Conservation analysis shows the ATP-binding pocket is highly conserved, suggesting inhibitor cross-reactivity is likely. Therapeutically, KIF11

inhibition has been targeted at anti-proliferative cancer treatments, but systemic toxicity and mitotic arrest limit its druggability in many cases.